

JEE MAIN REVISION MASTER: MAGNETIC PROPERTIES OF MATTER SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 73 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetic Properties Of Matter - Core Application	73 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Properties Of Matter - Core Application (Very High Yield — 73 PYQs)

Core Principle: Key problem pattern in Magnetic Properties Of Matter.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . $T \dots$ " [Ans: B]
- [2026 Morning Shift] Q#3 (MCQ): "Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material ..." [Ans: D]
- [2025 Morning Shift] Q#1 (MCQ): "The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibilit..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#2)

A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . The magnetic intensity in SI units is $\alpha \times 10^5$. The value of α is ____ . (Free space permeability $\mu_0 = 4\pi \times 10^{-7}$ SI units.)

- (A) $(25 / \pi)$
- (B) $(1 / 16\pi)$ **[Correct]**
- (C) $(1 / \pi)$
- (D) $(1 / 4\pi)$

Correct Answer: B

Solution Analysis: The relative permeability of the core is $\mu_r = 400$. The magnetic field applied is, $B = 1.0\text{T}$. The permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. The magnetic induction B in a material is related to the magnetic intensity H by the total permeability of the medium (μ) : $B = \mu H$... (i) The total permeability μ is the product of the permeability of free space and the relative permeability of the material : $\mu = \mu_0 \mu_r$ Substituting this into the first equation : $B = \mu_0 \mu_r H \Rightarrow H = \frac{B}{\mu_0 \mu_r}$ Substituting